

SS: $Q[\Phi^{\rm P}, \Phi^{\rm P}]$ closed form in 3D; modes continued analytically (the contour selects the finite part)

First-order residual $\Phi^{\rm res} = \Phi_1 - \Phi^{\rm P}$ on the real line (modes $\phi^{\rm res}_{\ell'}$, $\ell' \le \ell'_{\rm max}$)

Taylor polynomial about $\mathbf{x}_0$: $T_k\Phi^{\rm res} = \Phi^{\rm res}(\mathbf{x}_0) + \Delta x^i\, \partial_i \Phi^{\rm res}(\mathbf{x}_0) + \ldots$ (to order k), from mode sums at r_0 of the *same* truncated residual

RR: $Q[\Phi^{\rm res}, \Phi^{\rm res}]$ smooth at $\mathbf{x}_0$; solved on the real line throughout

subtract

SR Part 2: $2Q[\Phi^{\rm P}, R_k]$ $R_k = \Phi^{\rm res} - T_k\Phi^{\rm res} = O(\rho^{k+1})$ vanishes on the worldline, so the source is $O(\rho^{k-d})$: integrable, no δ terms, if $k \ge d-2$ (here $d = 2$, so $k \ge 0$). Formed on the real line, then $\tilde{s}(z) = \int_I s(t)\, \delta_{\rm ext}(z - t)\, dt$

SR Part 1: $2Q[\Phi^{\rm P}, T_k\Phi^{\rm res}]$ closed form in 3D; carries every non-integrable term, e.g. $\Phi^{\rm res}(\mathbf{x}_0)\, \partial_z^2(1/\rho)$, whose modes contain $\delta(r - r_0)$; continued analytically, so on the contour these become $\delta_{\rm ext}$ poles

Contour solves inside the worldtube (real line outside): source $\tilde{s}$ = SS, and separately $\tilde{s}$ = Part 1 + Part 2 for SR. No product of two shadowed factors, so no cross-term (b). Take Re at a_R.

Φ_2 outside the worldtube = $\Phi_2^{\rm SS} + \Phi_2^{\rm SR} + \Phi_2^{\rm RR}$ (linear superposition of the three solves)